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A discrete-continuous optimization approach for the design and operation of synchromodal transportation networks

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ABSTRACT

This paper presents a multi-objective mixed-integer programming problem for integrating specific characteristics of synchromodal transportation. The problem includes different objective functions including total transportation cost, travel time and CO_2 emissions while optimizing the proposed network structure. Traffic congestion, time-dependent vehicle speeds and vehicle filling ratios are considered and computational results for different illustrative cases are presented with real data from the Marmara Region of Turkey. The defined non-linear model is converted into linear form and solved by using a customized implementation of the ϵ-constraint method. Then, the sensitivity analysis of proposed mathematical models with pre-processing constraints is summarized for decision makers.

1. Introduction

The needs and systemic complexities resulting from globalizing activities and increased customer expectations have led to a fast growth of international freight transportation activities. Total inland transport increased to 2362 billion ton-kilometers in 2016 according to the aggregated EU transport performance data (EUROSTAT, 2018). However, road transportation accounts for 75.1% of this large volume (around 90% in developing countries including Turkey) (EEA, 2015), resulting in high CO_2 emissions, energy consumption and profit losses. Therefore, new and innovative approaches for planning, designing, constructing, operating, and maintaining transportation systems are required due to global concerns and increased awareness on environmental issues. Turkay (2008) examined environmentally conscious supply chain systems and indicated that transportation based on sustainability approaches should be a significant consideration in sustainable supply chain management. He argued that the development of multi-modal transportation systems would play an important role in reducing the impact of transportation activities on the environment, especially in terms of greenhouse gas (GHG) emissions. The implementation of this approach in the transportation processes cannot be achieved easily, and the establishment of a sustainable transportation system requires a well-defined framework that includes stakeholders from the private sector, public authorities, and research institutions. Although the concept of intermodal transportation includes some operational challenges, it has been recognized worldwide as a promising approach to reduce logistics costs. The European Commission (1997) defined intermodal transportation as "*the movement of goods in a loading unit, which uses successively several modes of transport without handling of the goods themselves in transshipment between the modes*". Intermodal transportation uses different types of transportation modes for a single shipment from one location to another, but the key characteristic of intermodal transportation is to handle only container-based units, not the goods themselves. This term is most often associated with international container traffic.

Many papers in the literature provide a general overview and definition of intermodal transportation problem. Bontekoning, Macharis, and Trip (2004) provided an extensive literature survey by considering intermodal transportation activities and indicated main research needs to boost intermodal transportation in the logistics industry. Macharis and Bontekoning (2004); Wu, Yin, and Lawphongpanich (2011) considered the importance of network designs in intermodal transportation. Groothedde, Ruijgrok, and Tavasszy (2005) and Parola and Sciomachen (2005) studied the network design of intermodal transportation and tried to obtain optimum locations for intermodal hubs. Roso, Woxenius, and Lumsden (2009) studied the dry port concept and tried to link sea and rail transportation by determining capacities and locations of port hinterlands. Macharis, Caris, Jourquin, and Pekin (2011) studied the service network design problem and focused on the cooperation of inland terminals and line bundling. Bektas, Chouman,

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and Crainic (2010) developed intermodal network design models on tactical levels considering bi-objective problems. SteadieSeifi, Dellaert, Nuijten, Van Woensel, and Raoufi (2014) summarized the papers on intermodal transportation in three main levels; strategic, tactical, and operational. Demir, Bektaş, and Laporte (2014) examined the green intermodal service network design problem by selecting robust routing plans. Resat and Turkay (2015) addressed the design and implementation of intermodal transportation concept in real-life cases. In addition to work on modeling, Meng, Wang, and Wang (2012); Wolsey and Nemhauser (2014) proposed solution methodologies with exact solution algorithms by relaxing and decomposing the models.

Although there are some successful implementations of intermodal transportation in the world, its potential to improve service levels, reduce costs and environmental impacts has not been fully exploited by current supply chains. Cargo volumes are mostly taken as container-based units and dynamic adaptation of planning processes throughout transportation operations are not considered. One of the main reasons behind inefficient road transportation systems is the low level of occupancy ratios (empty running) of medium and heavy-duty-vehicles. Also, according to the European Environmental Agency, when the vehicles are loaded, only half of the capacity was utilized (EEA, 2015). This means that most of the vehicles run with partial load and there are many relatively small volume shipments in the network due to customer pressure, insufficient infrastructure and capacity problems. Therefore, the new concept, synchromodal transportation, has appeared as an extension of intermodal transportation. It refers to transportation systems with dynamic adaptation of planning processes by incorporating with information regarding changes and disturbances occurring in transportation operations (Lucassen & Dogger, 2012; Van Riessen, 2013). Although intermodal transportation has been well discussed in the literature, synchromodal transportation is still a new topic for researchers and there are not many studies related with synchromodal transportation. Gorris et al. (2011); TNO (2012); Ham (2012); Mes and Iacob (2016) have introduced some theoretical definitions and potential benefits of synchromodal transportation in their studies. Behdani, Fan, Wiegmans, and Zuidwijk (2016) and Van der Burg (2012) have examined some industrial case studies in their master theses. However, their studies do not cover all characteristics of synchromodal transportation. Study of Farahani, Noble, Klein, and Enayati (2018) highlights the impact of reducing energy consumption and cost in synchromodal supply chain design and operation. Zhang and Pel (2016) develops a model that enables a comparative analysis of intermodal and synchromodal operations from economic, societal, and environmental perspectives. Li, Negenborn, and De Schutter (2017) investigates cooperative synchromodal freight transport planning among multiple intermodal freight transport operators in different and interconnected service networks.

As shown in Fig. 1, there are some differences between synchromodal transportation and other types of transportation systems.

The main idea behind synchromodal transportation is to minimize cost, time and environmental impact by integrating capacities of vehicles, transportation modes and customer demands. The study of Spectrum (2012) mentioned that synchromodal transportation fulfills greater flexibility in transportation mode choices, improves reliability, shortens lead-time in transport chains and increases utilization of road, rail and sea transportation. This leads to reduction of transportation cost and increase in freight volumes at the same time. This means that it cannot depend on a single objective and should be considered as a multi-objective optimization problem. Therefore, the concept of optimality is replaced with Pareto optimality (efficiency) to make decisions on the cost, time and environmental factors simultaneously. In general, there are two conventional methods to generate Pareto solutions for multi-objective optimization problems, which are the $\in$-constraint method and the weighted sum method. In this study, the $\in$-constraint method is used because the weighted sum method may sometimes miss some of the efficient solutions in some feasible regions (Mavrotas, 2009). In the $\in$-constraint method, the proper solution algorithm takes one of the objective functions as main objective and other objective functions are used as constraints in the problem. While finding the Pareto solutions, the model obtains an improved solution for one of the objective functions but deteriorates at least one of the other objective function values. Many researchers have proposed different solution methodologies to address multi-objective transportation problems. Steuer (1986), Miettinen and Mäkelä (1999), and Mavrotas and Diakoulaki (1998) generated an approximate representation of the set of Pareto solutions for a special type of linear multi-objective problems. Mavrotas and Florios (2013) proposed a new version of the $\in$-constraint method to find the set of exact solutions in multi-objective problems by adding slack variables into the objective functions that are taken as constraints and penalty term into the main objective function.

Apart from the multi-objective characteristic of the synchromodal transportation, tracking of real-time transportation times on different arcs is another important part of this problem that directly depends on traffic congestion (number of total passenger and freight vehicles on the roads) and quantity of carried goods on each vehicle. There are many studies to link vehicle speeds and their volume/capacity ratios in the literature. The Bureau of Public Roads (BPR) introduced the BPR equation (Manual T.A., 1964) fitted to freeway speed flow data given in the highway capacity manual in 1965 (Highway Capacity Manual, 1985). Akcelik (1996) and Dowling, Singh, and Cheng (1998) modified the BPR equation and proposed more complex speed-time estimation models. In addition, Wang and Meng (2012) and Demir et al. (2014) examined travel time models including uncertain waiting times by using stochastic approaches in port operations and intermodal networks, respectively. Jabali, Woensel, and de Kok (2012) introduced a model to analyze travel times and CO_2 emissions in time-dependent vehicle routing problems.

Synchronization between customers and suppliers, usage of different transportation modes at the same time and increase volume of freight flows by combining goods are the main essential assets of synchromodal transportation. After creating network system with these characteristics, synchromodal transportation may lead to

- better utilization of existing infrastructure,
- sustainable transport systems,
- reduction in CO_2 emissions,
- maximum usage of all transportation modes,
- increase in the position of sea and rail modes in transportation systems,
- better synchronization of customer demand and transportation operations,
- change in supply chain system from pull type to push type under coordination of control tower,
- reduced inventory levels by using more accurate forecast data (Pfoser, Treiblmaier, & Schauer, 2016).

The contribution of this paper is to introduce theoretical analysis of synchromodal transportation and to present novel models and solution methodologies to be used in real-life cases. The optimization models are analyzed in detail and various new pre-processing approaches were developed for the problem. It is shown that these pre-processing approaches are highly effective and proposed solution methodology leads to obtain Pareto solutions faster and decrease complexity of such large problems. The Pareto solutions obtained from the proposed models help stakeholders to use different transportation modes flexibly that enable better the utilization of existing infrastructure capacities.

This paper is divided into five sections. Section 2 presents the necessary technical background on the problem and provides a

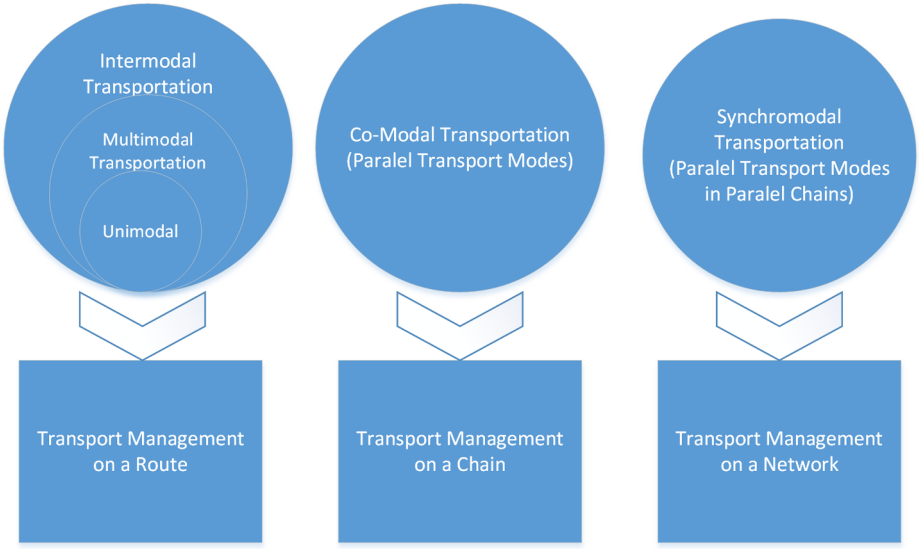


Fig. 1. Classification of different transportation types (BCI, 2011).

mathematical model for minimization of time, cost and CO_2 emissions factors in synchromodal transportation problem by considering traffic congestion. Section 3 includes information about the initial data and some parameters that are used during the calculations. Section 4 gives some information for solution methodology of this problem and results of the different illustrative cases. Finally, Section 5 shows the general overview and some future steps of the problem.

2. Problem statement

In this section, we define a synchromodal transportation-planning problem and present the formulation of both mixed-integer linear and non-linear optimization models for multi-objective problem that examine cost, time and CO_2 emissions functions for different transportation modes. While investigating the synchromodal transportation network design and operation, traffic congestion constraint is included into model with capacity constraints. In addition, CO_2 emissions rates are evaluated according to vehicle speeds and transportation durations.

2.1. Problem definition

The transportation industry involves many different players throughout its operations such as freight forwarders, customers, suppliers, and ports. Our problem is designed to simplify the logistics channels and provide the aspects in operational, tactical and financial levels. While describing the illustrative cases, some critical aspects for freight transportation given in the study of Cane (2012) are considered and listed in Table 1.

Table 1
Key performance indexes for freight transportation.

Area	Performance Indexes
Asset Management	Additional Vehicle Usage for Partial Cargoes
	Additional Travels of Vehicles
	Utilization of Investments
	Utilization of Freight Space
Customer Service	Order Fulfillment of Customers
	Backlog/Waiting Times in Orders
	Customer Satisfaction
Revenue Management	Annual Transportation Cost
	Additional Revenue due to the service improvements

After observing key performance indexes given in Table 1, illustrative cases are designed considering these needs. The problem starts with the data collection and prediction phase. In this part, historical traffic data, location of customers, information on the fleet (capacities, number per logistics providers, etc.) and initial data for customer orders that indicates the information related with transshipments, total cargo types and quantities are collected. By using all of this information, general time-dependent intermodal transportation model proposed by Resat and Turkay (2015) is used to decide on the transportation modes and quantities carried. The model uses the bundled quantity of customer orders as a total cargo volume in initial calculations. The main purpose of this model is to determine which transportation mode should be used and how much freight should be carried on the links of a given network. After determining the freight routes and transportation mode types, it should be visible for everyone to trace and track their cargoes. Information on the arrival and departure times of vehicles is one of the main aspects of the synchromodal transportation: after obtaining the results of intermodal transportation model, the fleet optimization method is used to determine the optimal usage of the fleet for each transportation mode. It is important to increase the fill ratios of the fleet and determine the number of trucks in road transportation because it directly affects the traffic congestion and deteriorate the transportation times and CO_2 emissions. Although the total freight amount is assessed as a unit of 40-feet container (20 ton) in intermodal transportation problems, this is not realistic in real-life medium or heavy-duty trucks carry approximately 77% of the total cargo in road transportation and containers are not used in such operations (EPA, 2013). The number of vehicles required can be obtained in this problem and this information directly related with traffic congestion is used by time-dependent synchromodal transportation model (TDSTM) to obtain the optimum results of the problem. We can obtain network structure and locations of the vehicles from TDSTM. However, our model has a dynamic approach and it starts up its solution methodology after finding the minimum transport duration from TDSTM. After finding the minimum transportation duration, we can trace the status of vehicles in the network and we can assume that there will be some unloading operations in one of the vehicles of the fleet. This means that if there is some free space for new cargo ordered by the customer, our optimization model will assess the new circumstances and supply new cargo for the TDSTM. If there is no free space on the truck or new cargo is not available, our model uses the previous results and continues until the next transportation period. After this operation, this dynamic problem

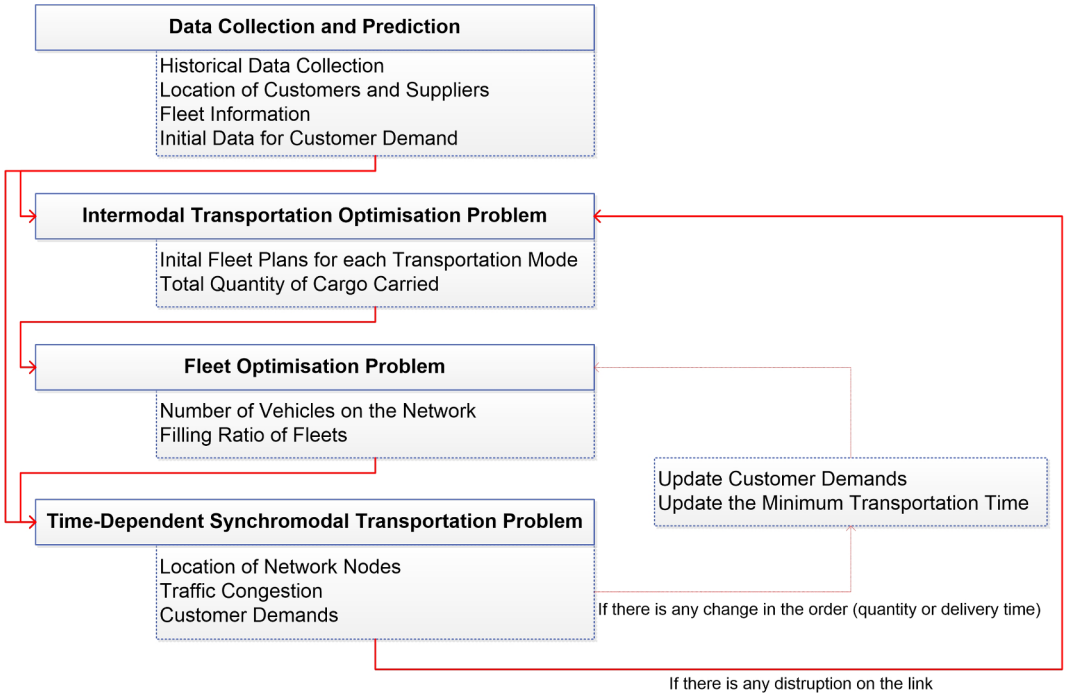


Fig. 2. A flowchart of the solution methodology.

is solved again by checking the new orders and taking decisions over the process.

In addition, there are usually disruptions during the operations in real-life cases. In such cases, our problem can assess this condition and start the process with new input from the intermodal optimization problem and decides on transportation modes and network for the disrupted link only. After deciding on the transportation mode and route, fleet optimization model is used to optimize the number of vehicles in the network. It can either combine that cargo if there is enough free space in some vehicles or order new vehicles to carry that cargo in determined transportation mode (see Fig. 2).

2.2. Model formulation

In this section, we present two models for TDSTM: multi-objective mixed-integer linear programming (MOMILP) and non-linear programming (MOMINLP). The main objectives involve the total transportation cost, the total time spent on the system and the total CO_2 emissions amounts. Our model is based on indices that are indicated below:

The model contains six sets: set of nodes, $C = \{(i,j)|i,j \in C\}$, set of transportation modes, $K = \{(k,l)|k,l \in K\}$, set of time intervals, $T = \{(t \in T\}$, set of vehicles, $V = \{v \in V\}$, set of vehicle speed levels, $S = \{s \in S\}$ and set of regions of speed module based on congestion, $R = \{m \in R\}$. Lets define W as the set of all nodes $W = C \cup K \cup T \cup V \cup S \cup R$. According to the paper of Bektaş & Laporte (2011), the planning horizon is divided into three sections that are all congestion, transient, all free-flow regions. In these different time modules, vehicles are allowed to move during high traffic congestion period or mixture of free-flow and congestion or free-flow periods (no traffic jam) respectively. For example, if a vehicle leaves one of the nodes in the congestion module ($m = 1$), this means that the vehicle moves all the way on arc (i,j) with the minimum speed levels due to traffic congestion and it will take much more time. However, if it moves in the transient period ($m = 2$), it can either start with traffic jam but after some time traffic may disappear and it can move faster or vice versa. In the last module ($m = 3$), vehicle moves without any traffic congestion and it can reach the targeted node the fastest.

We assume that all of the initial data for demand of each node for different transportation modes at different time intervals (D_{ikt}); locations and distances between these nodes (d_{ijk}); operational data such as opening (Δ_{ikt}) and closing of the nodes (μ_{ikt}); loading of vehicles in each node for different transportation modes (L_{ikv}); capacity of the nodes for each transportation modes (C_{ik}); number of vehicles and their capacities (Cap_v^{max}) are given beforehand. Therefore, the process starts from the determined logistics centers (nodes) and conventional vehicles are used to transport the bulk or partial cargoes between the customers with a unit transportation costs per transportation modes (TC_{ijk}). Sometimes, there are changes between the transportation modes at the nodes and this operation leads to additional transfer cost (TrC_{kl}) and transfer time (TrT_{kl}) which also take a fixed value for each transportation mode and node. This operation requires the decision on opening links between nodes and how much cargo and which transportation mode is used for cargo carriage on the arcs defined as $A = \{(i,j)|i,j \in C,\ i \neq j\}$. In addition, the solution of the problem identifies the penalty costs if there is any waiting process due to the early or late delivery to the customers that are indicated as P^e, P^g and P^d for early, late arrival and delay in normal operations respectively. The penalty of the waiting times is designed according to the real-life cases. Therefore, the values for P^e and P^g are much larger than P^d because these two options are undesired operations for just-in-time real-life cases and model tries to minimize waiting times due to late and early arrivals. The value of P^g (for late arrivals) is a bit larger than the P^e because the vehicles wait more in the case of late arrivals in general (next day deliveries). In addition, we penalize normal operations during the daytime that take longer than expected. The value of P^d is smaller than the amounts for P^e and P^g. The model penalize keeping inventory in node i by unit inventory cost of P_{inv}.

In addition, the model includes the effect of the environmental aspects by calculating the CO_2 emissions in terms of speed levels and other parameters for this purpose (Franceschetti, Honhon, Van Woensel, Bektaş, & Laporte, 2013).

S_k^{ns}	Speed limit of transportation mode k for free-flow transportation
b_{ijkvm}	Time constant of corresponding travel time module m for vehicles v traveling on arc (i,j) by transportation mode k

$$b_{ijkvm} = \begin{cases} 0 & m = 0 \\ \left(alfa - \frac{d_{ijk}}{S_k^s}\right) & m = 1 \\ alfa & m = 2 \end{cases}$$

θ_{ijkvms}	Carbon dioxide emissions constant

$$\theta_{ijkvms} = \begin{cases} 0 & m = 1,\ 3 \\ \frac{S_k^{2s} - S_k^{3s}}{S_k^{3s}} & m = 2 \end{cases}$$

ω_{ijkvms}	Carbon dioxide emissions constant

$$\omega_{ijkvms} = \begin{cases} 0 & m = 0 \\ \frac{d_{ijk}}{S_k^{1s}} & m = 1 \\ \frac{d_{ijk}}{S_k^{1s}} + \left(\frac{S_k^{3s} - S_k^{2s}}{S_k^{3s}}\right) & m = 2 \\ \frac{d_{ijk}}{S_k^{3s}} & m = 3 \end{cases}$$

In addition, there are some parameters in the optimization of the multi-objective problem.

G_{time}	Any large value greater than upper bound of the secondary objective function
G_{CO2}	Any large value greater than upper bound of the third objective function

There are many scalars for this problem and they are mostly used in speed and carbon dioxide calculations. In addition, when the problem is designed as a linear problem, some large values (known as big-M parameters) are used in linearization process.

Scalars:

a, b	BPR equation constants
A,B	Carbon dioxide emissions rate constants
α, β, π, λ	Carbon dioxide emissions rate constants for non-linear model
$\in$	Epsilon value
σ	Curb weight (kg)
Ω	Daily period (hours)
h	Engine Friction Factor constant
U	Engine Speed constant
H	Engine Displacement constant
$M1$, $M2$, $M3$, $M4$	The big-M parameter values

There are three main objectives in this problem; cost, time and CO_2 emissions. While the problem takes the cost objective as a priority action and tries to minimize the cost variables, the time and environmental objectives are also taken into account. Therefore, the problem initially minimizes the total number of carried goods (X_{ijkvt}) on arc (i,j) and total number of inventories (I_{ikt}) for each transportation mode in different time intervals. This means that the model minimizes the number of the vehicles on each arc (i,j) and this amount is directly related with the capacity fill ratios of the vehicles because the capacity of vehicles are taken as constant for each transportation mode. If the vehicles are filled up to their maximum capacity (mostly in road transportation) and carry the maximum amount of goods (X_{ijkvt}) on arc (i,j), this directly affects the number of the vehicles on the arcs. Therefore, the total transportation cost can be minimized. Also, if there is any transfer between the transportation modes, then this creates some additional cost for the system and the model tries to reduce this cost variable. However, minimization of the cost alone is not enough, other objectives must also be optimized. Therefore, the model considers the total transportation duration (TT_{ijkvt}) on arcs (i,j) and also time consumption during the transfer operations. Additionally, the model controls the operational movements of the cargo in terms of their time frames, such as the arrival (ta_{ijkvt}) and departure times (td_{ijkvt}) of the vehicles to/from the nodes for different transportation modes; the waiting times for the vehicles in each node for different transportation modes (EW_{ikvt}, LW_{ikvt}, NW_{ikvt} for early, late and normal arrival cases respectively). Moreover, the model provides information on the fill ratio of the vehicles (γ_{vkt}) for different transportation modes in different time intervals and time instant (Λ_{ijkvms}) at which vehicle v departs node i in time interval m by mode k and speed s. In addition, there are some binary variables to control the transportation operations effectively.

Binary Variables:

y_{ijkvt}	$\begin{cases} 1 & \text{if transportation mode } k \text{ is chosen on arc } (i,j) \text{ in time interval } t \\ 0 & \text{else} \end{cases}$
w_{iklt}	$\begin{cases} 1 & \text{if transfer from transportation mode } k \text{ to } l \text{ in node } i \text{ in time interval } t \\ 0 & \text{else} \end{cases}$
z_{ijkvt}^{e}	$\begin{cases} 1 & \text{if vehicle } v \text{ in node } i \text{ for transportation mode } k \text{ in period } t \text{ arrives early} \\ 0 & \text{else} \end{cases}$
z_{ijkvt}^{g}	$\begin{cases} 1 & \text{if vehicle } v \text{ in node } i \text{ for transportation mode } k \text{ in period } t \text{ arrives late} \\ 0 & \text{else} \end{cases}$
z_{ijkvt}^{d}	$\begin{cases} 1 & \text{if vehicle } v \text{ in node } i \text{ for transportation mode } k \text{ in period } t \text{ arrives in time} \\ 0 & \text{else} \end{cases}$
ψ_{ijkvms}	$\begin{cases} 1 & \text{if vehicle } v \text{ departs node } i \text{ in time interval } m \text{ by mode } k \text{ and speed } s \\ 0 & \text{else} \end{cases}$

First, the non-linear model is developed with the main objective functions and constraints as given below:

$$\boldsymbol{min} \quad f_1 = \sum_{t\in T}\sum_{v\in V}\sum_{i\in I}\sum_{j\in J}\sum_{k\in K}(TC_{ijk}\times X_{ijkvt}\times y_{ijkvt}) + \sum_{t\in T}\sum_{i\in I}\sum_{k\in K}\sum_{l\in L}(TrC_{kl}\times w_{iklt}) + \sum_{i\in I}\sum_{t\in T}\sum_{k\in K}(I_{ikt}\times P_{inv}) \tag{1}$$

$$\boldsymbol{min} \quad f_2 = \sum_{t\in T}\sum_{v\in V}\sum_{i\in I}\sum_{j\in J}\sum_{k\in K}(TT_{ijkvt}\times y_{ijkvt}) + \sum_{t\in T}\sum_{i\in I}\sum_{k\in K}\sum_{l\in L}(TrT_{kl}\times w_{iklt}) + \sum_{i\in I}\sum_{t\in T}\sum_{k\in K}\sum_{v\in V}\{((EW_{ikvt}\times P^e + LW_{ikvt}\times P^g) + (NW_{ikvt}\times P^d)\} \tag{2}$$

$$\boldsymbol{min} \quad f_3 = \sum_{t\in T}\sum_{v\in V}\sum_{i\in I}\sum_{j\in J}\sum_{k\in K}\lambda\left(hUH + \sigma\pi\alpha\frac{d_{ij}}{TT_{ijkvt}} + X_{ijkvt}\pi\alpha\frac{d_{ij}}{TT_{ijkvt}} + \beta\pi\left(\frac{d_{ij}}{TT_{ijkvt}}\right)^3\right)\times TT_{ijkvt} \tag{3}$$

$$EW_{ikvt} \geq [\Delta_{ikt} - t\alpha_{ijkvt}] + M3(1 - z^e_{ijkvt}) \qquad t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{4}$$

$$LW_{ikvt} \geq [\Omega - t\alpha_{ijkvth} + \Delta_{ik(t+1)}] + M3(1 - z^g_{ijkvt}) \qquad \forall\ t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{5}$$

$$NW_{ikvt} \geq [td_{ijkvth} - ta_{ijkvt}] + M3(1 - z^d_{ijkvt}) \qquad \forall\ t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{6}$$

$$z^e_{ijkvt} + z^g_{ijkvt} + z^d_{ijkvt} = y_{ijkvt} \qquad \forall\ t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{7}$$

$$td_{ijkvt} \geq ta_{ijkvt} + L_{ikv} + TrT_{kl} - M4(1 - y_{ijkvt}) \qquad \forall\ t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{8}$$

$$\Delta_{ikt} \leq td_{ijkvt} \leq \mu_{ikt} \qquad \forall\ t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{9}$$

$$\sum_{j\in J} y_{ijkvt} \geq 0 \qquad \forall\ t\in T,\ i\in C,\ k\in K,\ v\in V \tag{10}$$

$$I_{ikt} \leq C_{ik} \qquad \forall\ t\in T,\ i\in C,\ k\in K \tag{11}$$

$$\sum_{l\in L} w_{iklt} \geq 1 \qquad \forall\ t\in T,\ i\in C,\ k\in K \tag{12}$$

$$y_{jikvt} + y_{ijkvt} \geq 2\times w_{iklt} \qquad \forall\ t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{13}$$

$$I_{ik(t-1)} + \sum_{t'\in T}\sum_{v\in V}\sum_{j\in J} X_{jikvt'} = \sum_{v\in V}\sum_{j\in J} X_{ijkvt} + I_{ikt} \qquad \forall\ t\in T,\ i\in C,\ k\in K \tag{14}$$

$$t - 1 \leq t' + TT_{jikvt'h'} \leq t \qquad \forall\ t'\in T \tag{15}$$

$$\sum_{v\in V}\sum_{j\in J} X_{jikvt} - \sum_{v\in V}\sum_{j\in J} X_{ijkvt} = D_{ikt} \qquad \forall\ t\in T,\ i\in C,\ k\in K \tag{16}$$

$$D_{ikt}\times y_{ijkvt} \leq X_{ijkvt} \leq (Cap_v^{max} - D_{ikt})\times y_{ijkvt} \qquad \forall\ t\in T,\ (i,j)\in C,\ k\in K,\ v\in V \tag{17}$$

$$TT_{ijkvt} = \frac{d_{ijk}}{S_k^{ms}}\times\left[1 + a\times\left[\frac{X_{ijkvt}}{C_{ik}}\right]^b\right] \qquad \forall\ t\in T,\ (i,j)\in A,\ m\in M,\ s\in S,\ v\in V,\ k\in K \tag{18}$$

$$\gamma_{vkt} = \frac{\sum_{i\in I}\sum_{j\in J} X_{ijkvt}}{Cap_v^{max}} \qquad \forall\ t\in T,\ v\in V,\ k\in K \tag{19}$$

$$\gamma_{vkt} \leq 1 \qquad \forall\ t\in T,\ v\in V,\ k\in K \tag{20}$$

$$X_{ijkvt},\ Q_{ijkvt},\ J_{ijkvt},\ TT_{ijkvt},ta_{ijkvt},\ \Lambda_{ijkmvs} \geq 0 \qquad \forall\ t\in T,\ (i,j)\in C,\ v\in V,\ k\in K,\ s\in S \tag{21}$$

$$z^e_{ijkvt},\ z^g_{ijkvt},\ z^d_{ijkvt} \geq 0 \qquad \forall\ t\in T,\ (i,j)\in C,\ v\in V,\ k\in K \tag{22}$$

$$EW_{ikvt},\ LW_{ikvt},\ NW_{ikvt} \geq 0 \qquad \forall\ t\in T,\ i\in C,\ v\in V,\ k\in K \tag{23}$$

$$y_{ijkvt},\ w_{iklt},\in \{0,\ 1\} \qquad \forall\ t\in T,\ (i,j)\in A,\ v\in V,\ k\in K \tag{24}$$

$$\psi_{ijkvms}\in\{0,\ 1\} \qquad \forall\ t\in T,\ (i,j)\in C,\ m\in R,\ s\in S,\ v\in V,\ k\in K \tag{25}$$

$$I_{ikt} \geq 0 \qquad \forall\ t\in T,\ i\in C,\ k\in K \tag{26}$$

Eqs. (1)–(3) model the main objective functions of this model, which are cost, time and CO_2 emissions rate respectively. In Eq. (1), the first term corresponds to the total transportation cost of carried goods, the second term gives the transfer cost among transportation modes and the last one is the penalty cost of keeping inventory in nodes. Eq. (2) shows that the minimization of the total transportation durations during the operations and the total transfer time in mode changes and the third term of this equation gives the penalty to minimize the waiting times of vehicles in nodes. In Eq. (3) indicates the minimization of the total CO_2 emissions in the system. The first term of this equation gives the emission caused by the engine module and the speed module, the second term shows total the emission generated by the speed module in the all congestion and in the all free-flow regions, and the last one represents the total emission induced by the weight module (Bektaş & Laporte, 2011; Demir et al., 2014). Eqs. (4)–(6) indicate that the total waiting time of vehicles for each transportation mode; there should be early, late or in time arrival, probabilities in the allowed time-period and Eqs. (4)–(6) fulfill this situation respectively. Eq. (4) gives the waiting time of the vehicles due to early arrival to node *i* which means that the total waiting period between opening time of the logistics center in node *i* and arrival time of the vehicle to the node *i*. Eq. (5) gives the waiting time due to the late arrival of the vehicles. This equation indicates that the vehicle arrives to the node *i* after the closing time of the logistic center and it should wait until the opening time of the center next morning. Eq. (6) shows the total spent time in node *i* for normal operations; the main purpose of this equation is to minimize the total spent time or delays in the nodes and reduce unnecessary time losses. E q. (7) ensures that if there is any freight to be carried on arc *(i,j)*, each vehicle in the nodes for different transportation modes has to arrive early, late or in time. Eq. (8) indicates the relationship between the departure and arrival times of vehicles within nodes for different transportation modes. The duration between the arrival and departure times of each vehicle has to be greater than the summation of loading time of the vehicle for mode *k* and transfer times from mode *k* to *l*. Eq. (9) ensures that the departure time of vehicle has to be between the opening and the closing time of nodes for each transportation mode. For example, if the sum of loading and transfer time between different transportation modes exceed the closing time of the node for a particular transportation mode, then the vehicle has to wait until the next day and quantity on vehicle will be added to inventory of node for that transportation mode. Eq. (11) indicates that the daily inventory of every node *i* is restricted by the capacity of each transportation mode *k*. Eq. (12) indicates that any material in node *i* must be transported via at least one of the available transportation modes. This constraint forces the solver to obtain the minimum transfer cost and transfer time in node *i* in terms of the transportation modes. Eq. (13) shows that the arrival transportation mode must be the same transportation mode of the leaving mode over the arc *(i,j)*. On the other hand, if there is any difference between arriving and leaving transportation modes, Eq. (13) ensures that there must be a mode change within the node. Eq. (14) indicates the mass balance in the nodes; the incoming and outgoing quantities in node *i* for each transportation mode *k* should be same, and Eq. (14) helps model to fulfill this balance during the operations. However, the relationship between current (*t*) and previous time intervals ($t' < t$) has to be as in Eq. (15). The arrival time of vehicles on

arc (*i,j*) for different transportation modes, which is equal to the sum of departure time of vehicles on arc (*i,j*) and transportation duration on arc (*i,j*) for different transportation modes, has to be in the range of single time period (between t and $(t-1)$). Travel times are taken as a step-wise function for 24-h period per day in this part of the problem. Eq. (16) shows that incoming and outgoing quantities of carried goods have to be equal to demand of nodes for different transportation modes in every period. Capacities of vehicles are satisfied by Eq. (17) for each transportation mode in every period. Eq. (18), the BPR (Bureau of Public Roads) equation (1964) (Manual T.A., 1964) is used to model the traffic congestion. The BPR equation was modified to obtain transportation times of the modes for every time interval *t*. The BPR equation constants were obtained from Horowitz (1985). Eqs. (19) and (20) indicate the occupancy rate of vehicles for different transportation modes. Eqs. (21)–(26) ensure both integrality and non-negativity conditions.

The proposed model focuses on three main objectives; cost, time and CO_2 emissions rate. As given in the Table 1, some critical aspects should be considered in transportation management problems. For example, one of the main objectives is directly related with the revenue management section (cost). The model tries to minimize the total transportation cost in the given network, decreases the total number of vehicles in the network and this reduces the need for additional vehicles in partial cargoes. In addition, combining the cargoes in single vehicle leads to the reduction of additional travel of cargo and increases the utilization of the vehicles. Secondly, the model can satisfy the customer expectations in terms of waiting times and order fulfillment by using the second objective of the model (time). The problem gives Pareto solutions to send cargoes in minimum time and cost by considering the environmental effects. In addition, the decision makers should assess the system in terms of the investments using information on traffic densities, inventory levels and waiting times of the vehicles from the results of the problem. For example, there are always waiting queues or higher inventory levels at logistics centers, and then decision makers may take some actions to address these problematic cases.

2.3. Solution methodology

In this mathematical model, three are three objective functions that are cost minimization (f_1), time minimization (f_2) and CO_2 emissions minimization (f_3). The augmented ϵ-constraint method is used to generate the Pareto optimal solutions for the decision makers in this study.

Cost minimization is always more important than other objectives for real-life cases due to harsh competitive conditions. Therefore, f_1 is selected as a prior function and f_2 and f_3 are taken as secondary and third objectives and inserted as constraints in the augmentedϵ-constraint method. The multiplication of positive continuous slack variables ($X^1_{slack}$and X^2_{slack}) with some small constant value ($\epsilon = 0.00001$) is subtracted from the objective function, however same slack variables are included into f_2 and f_3 at the same time. Summation of these slack variables with f_2 and f_3 should be equal to the corresponding upper bounds of the f_2 and f_3.

$$Min \quad f_1 - \epsilon * (X^1_{slack} + X^2_{slack}) \tag{27}$$

Subject to:

$$f_2 + X^1_{slack} = G_{time} \tag{28}$$

$$f_3 + X^2_{slack} = G_{CO2} \tag{29}$$

$$X^1_{slack}, X^2_{slack} \geq 0 \tag{30}$$

Eqs. (4)–(26)

The relationship between objective functions are determined and applied for different problem cases. However, finding the optimal solutions for this type of non-linear problems has high computational complexity due to the large number of constraints and indices that requires serious computing resources (Mahlke, Martin, & Moritz, 2010). Therefore, we consider the linearization of this model that gives rise to a multi-objective mixed-integer linear programming problem.

2.3.1. Linearization of the model

$$\textbf{min} \quad f_1 = \sum_{t\epsilon T}\sum_{v\epsilon V}\sum_{i\epsilon I}\sum_{j\epsilon J}\sum_{k\epsilon K}(TC_{ijk} \times X_{ijkvt}) + \sum_{t\epsilon T}\sum_{i\epsilon I}\sum_{k\epsilon K}\sum_{l\epsilon L}(TrC_{kl} \times w_{iklt}) + \sum_{i\epsilon I}\sum_{t\epsilon T}\sum_{k\epsilon K}(I_{ikt} \times P_{inv}) \tag{31}$$

$$\textbf{min} \quad f_2 = \sum_{t\epsilon T}\sum_{v\epsilon V}\sum_{i\epsilon I}\sum_{j\epsilon J}\sum_{k\epsilon K}(J_{ijkvt}) + \sum_{t\epsilon T}\sum_{i\epsilon I}\sum_{k\epsilon K}\sum_{l\epsilon L}(TrT_{kl} \times w_{iklt}) + \sum_{i\epsilon I}\sum_{t\epsilon T}\sum_{k\epsilon K}\sum_{v\epsilon V}\{((EW_{ikvt} \times P^e + LW_{ikvt} \times P^g) + (NW_{ikvt} \times P^d)\} \tag{32}$$

$$X_{ijkvt} \leq Q_{ijkvt} \qquad \forall\, t \in T, (i,j) \in A, k \in K \tag{33}$$

$$X_{ijkvt} \leq M \times y_{ijkvt} \qquad \forall\, t \in T, (i,j) \in A, k \in K \tag{34}$$

$$X_{ijkvt} \geq Q_{ijkvt} - M(1 - y_{ijkvt}) \qquad \forall\, t \in T, (i,j) \in A, k \in K \tag{35}$$

$$J_{ijkvt} \leq TT_{ijkvt} \qquad \forall\, t \in T, (i,j) \in A, k \in K \tag{36}$$

$$J_{ijkvt} \leq M \times y_{ijkvt} \qquad \forall\, t \in T, (i,j) \in A, k \in K \tag{37}$$

$$J_{ijkvt} \geq TT_{ijkvt} - M(1 - y_{ijkvt}) \qquad \forall\, t \in T, (i,j) \in A, k \in K \tag{38}$$

Eqs. (33)–(38) indicate the linearization of the model because the multiplication of carried quantity on arc (*i,j*) for each transportation mode and the binary variable giving decision whether using that arc (*i,j*) by the same transportation mode makes the model non-linear and model can be converted into a linear one (Eqs. (33)–(35)). Same situation is also valid for transportation time calculations. The multiplication of two variables creates non-linear conditions and relaxation of these equations by using the Eqs. (36)–(38) convert our problem into a linear form.

Also, BPR equation (Manual T.A., 1964), Eq. (18), makes the problem non-linear because the transportation time is directly related with the number of vehicles on arc (*i,j*). In the conventional model, the constant b of BPR equation is taken as "4" and this makes the model non-linear, but study of Resat and Turkay (2015) indicates that constant b can be taken as "1" to handle the non-linearity of the Eq. (18) and this change creates approximately less than 5% difference in the results and the effect of this change can be negligible.

Additionally, CO_2 emissions rate calculation is the one of our objectives and this factor directly related with the fuel consumption, same as total travelled distance by the vehicles and the equation used in this model makes the problem non-linear. However, we replace this objective function with the linear one developed by the recent studies of Bektaş & Laporte (2011). It combines the speed of the vehicles with the CO_2 emissions rate in terms of the time dependency. The equations are given below.

$$\min \quad f_3 = \sum_{m\in M}\sum_{i\in I}\sum_{j\in J}\sum_{s\in S}\sum_{k\in K}\{\{(\theta_{ijkms}\times\Lambda_{ijkms}+\omega_{ijkms}\times\psi_{ijkms})\times A + (\theta_{ijkms}\times\Lambda_{ijkms}+\omega_{ijkms}\times\psi_{ijkms})\times(S_k^{ms})^3\times B\}\} + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K}\sum_{s\in S}\{\{(\Lambda_{ijk2s}+\theta_{ijk2s}\times\Lambda_{ijk2s}+\omega_{ijk2s}\times\psi_{ijk2s} - \alpha\times\psi_{ijk2s})\times(S_k^{3s})^3\times B\} + \{(\alpha\times\psi_{ijk2s}-\Lambda_{ijk2s})\times(S_k^{2s})^3 \times B\}\} \tag{39}$$

$$\psi_{ijkvms}\times b_{ijkv(m-1)} \le \Lambda_{ijkvms} \le \psi_{ijkvms}\times b_{ijkvm} \qquad \forall (i,j)\in A,\ m\in M,\ k\in K,\ v\in V,\ s\in S \tag{40}$$

$$\sum_{m\in M}\sum_{s\in S}(\theta_{ijkvms}\times\psi_{ijkvms}+\omega_{ijkvms}\times\Lambda_{ijkvms}) \le \sum_{h\in H} TT_{ijkvt} \qquad \forall t\in T,(i,j)\in A,\ v\in V,\ k\in K \tag{41}$$

Eq. (40) ensures the period at which vehicle v leaves the node i by transportation mode k under boundary limits. Eq. (41) indicates that the ravel time (of which vehicle v on arc (i,j) by transportation mode k) used in the CO_2 emissions calculations is at most equal to transportation time of speed calculations.

After converting the problem into the linear version, the problem still requires serious computational effort and reduction of the complexity of linear integer problems is the major area of the recent studies. In literature, heuristics, meta–heuristics and hybrid genetic algorithms are proposed (Ishfaq & Sox, 2012; Limbourg & Jourquin, 2009; Mahlke et al., 2010). However, this exact method is used to eliminate the poor accuracy of heuristics methods in the study of Resat and Turkay (2015). Especially, pre-processing and logic-cut methods help solvers to obtain more accurate and optimal solutions by decreasing the complexity and decomposition of the problem. Although the logic-cuts increase the number of constraints, it would reduce the relaxation gap and would strengthen the convex hull approximation. Therefore, solvers will be able to obtain faster in shorter CPU times. Also, addition of the extra constraints involving logic-cut and pre-processing methods does not change the integer optimal solutions. As indicated in the study of Fattahi and Turkay (2014), some logic-cuts and pre-processing methods help find solutions faster.

2.3.2. Pre-Processing

After modeling our problem in the linear form, we introduced some pre-processing steps and valid inequalities to increase the efficiency and to decrease the dimension of the problem.

$$y_{ijkvt} = 0 \qquad \forall t\in T,(i,j)\in A,\ k\in K,\ v\in V \qquad \text{if } (i,j) \text{ isn't linked} \tag{42}$$

$$\sum_{j\in J} y_{ijkt} = 0 \qquad \forall t\in T,\ i\in I,\ k\in K,\ v\in V \qquad \text{if } D_{ikvt}=0 \tag{43}$$

First, all cost matrices of transportation modes, transfer operations and distances between nodes are initialized as indicated in Eqs. (42) and (43). With Eqs. (42) and (43), the set of arcs (i,j) *are assessed* to see whether there is a link between these two nodes or there is sufficient demand of node i. If they are not connected or there is no demand in the origin node i, it is not possible to transport any goods from node i to node j.

The pre-processing steps aim to remove infeasible arcs from the problem as described below:

$$D_{ikt} + D_{jkt} \ge |V|_i \cdot Cap_v^{max} \qquad \forall t\in T,(i,j)\in A,\ k\in K \tag{44}$$

$$\Delta_{ikt} + L_{ikv} + J_{ijkvt} \ge \mu_{jkt} \qquad \forall t\in T,(i,j)\in A,\ k\in K,\ v\in V \tag{45}$$

Eq. (44) ensures that if the demand of the subsequent nodes exceeds the total capacity of the vehicles for each transportation mode in every time interval, then the system eliminates this arc. Eq. (45) indicates that if the delivery is later than the closing time of the following node, then this arc will be eliminated. Also, in our problem, time-dependent travel times are the step-wise function for 24 h time period per day. This quantity balance in Eq. (14) has to be satisfied for every time interval t. However, the relationship between t and previous time intervals ($t' < t$) has to be as in Eq. (15). To calculate the incoming quantities up to time interval t, travel duration of vehicles plays a crucial role as traffic congestion affects the travel duration of vehicles. To obtain the value of $\sum_{t'\in T}\sum_{j\in J} X_{jikvt'}$(incoming goods) in Eq. (14), the following logic system was created for time interval t > 2.

The transportation time matrix is constructed using Eq. (18). Then, the new parameter ($arrival_time_{jikt'}$) is defined that recognizes and keeps the arrival times of vehicles which leaves from node j at t' and reach to node i before time interval t. This transportation time matrix is constructed for road transportation only by using Eq. (46) because transportation times for sea and rail transportations are assumed to be constant and they are calculated by dividing the total distance on arc (i,j) by the average speed of the vehicles.

$$t' + TT_{jikvt'} = arrival_time_{jikvt'} \qquad \forall t'\in T,(i,j)\in A \qquad \text{for } k = Road \tag{46}$$

According to the arrival time values, a new binary matrix (AT) is used to indicate the whether there are any flows which depart from node j at t' and arrive at node i at t (equal to $arrival_time_{jikt'}$) for transportation mode k.

$$AT_{jikvt't} = \begin{cases} 1 & \text{if } arrival_time_{jikvt'} > 0 \\ 0 & \text{else} \end{cases} \qquad \forall t\in T,\ t'\in T,(i,j)\in A,\ k\in K \tag{47}$$

Then, this new parameter (time matrix AT) is multiplied with $X_{ijkt'}$ to obtain the new quantities regarding the transportation times. The total amount of incoming goods to node i by transportation mode k until time interval t-1 and t is modeled in Eq. (48).

$$\sum_{j\in J}\sum_{t'\in T} AT_{jikvt't}X_{jikvt'} \qquad \forall t\in T,\ i\in I,\ k\in K \tag{48}$$

3. Computational experiments

In this section, we present the parameters used (derived from the real data in the Marmara region of Turkey) in multi-objective optimization model, overview of the illustrative cases and their applications. Additionally, the assumptions used in the design and the operations of the synchromodal system are discussed.

First, we start our work with data collection and parameter prediction. For this purpose, we collected historical traffic data, the location of customers, information on the fleet (vehicle capacities, number of vehicles for each logistics provider, etc.) and the initial data for customer orders (their coordinates, total cargo type and quantities). At this stage, we used the benchmark data given in Resat and Turkay (2015), which used real-case data for the Marmara Region in Turkey. For the illustrative cases, we considered eight different combinations of available nodes that are the most important ports and logistics centers in the Marmara Region. After selecting the nodes for our illustrative cases, we gather data on the distances between the nodes; unit costs for transportation and transfer activities in the nodes, and average transfer times of cargo from official reports that were issued by the Ministry of Transportation and some public bodies of Turkey (DTM, 2010; KGM, 2011; TCDD, 2010). Resat and Turkay (2015) showed that 98% of the freight transportation activities are realized in combination with road, rail and sea transportation. The speed of vehicles in the road transportation (especially heavy trucks) is more sensitive and dependent on traffic congestion compared to the other two transportation modes. We took into consideration only these transportation modes. The speeds of the rail and sea transportation are taken as constant and the speed of road transportation is calculated with the BPR equation where the vehicle speed depends on the number of vehicles on the road. Therefore, it is assumed that only road transportation has variable travel times that

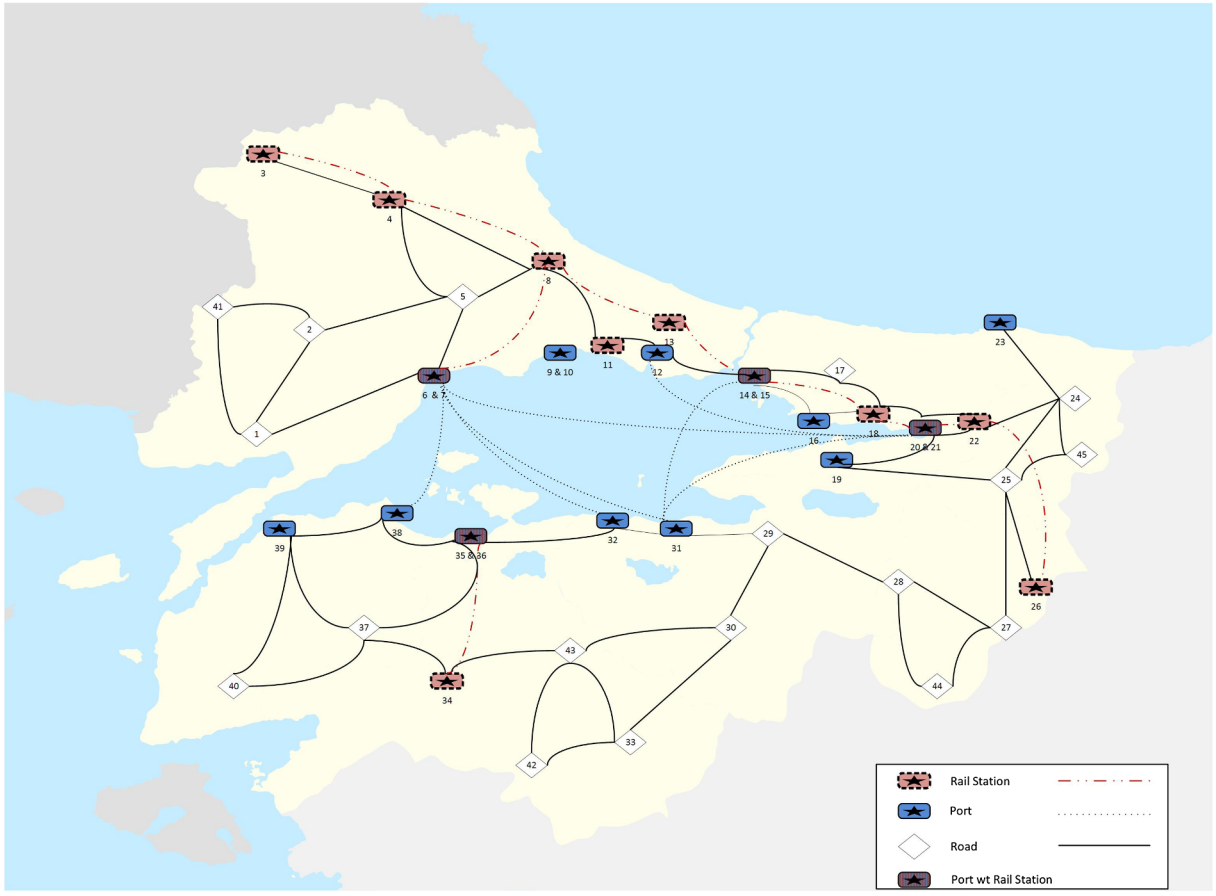


Fig. 3. Network design of the 45-node case with different transportation modes.

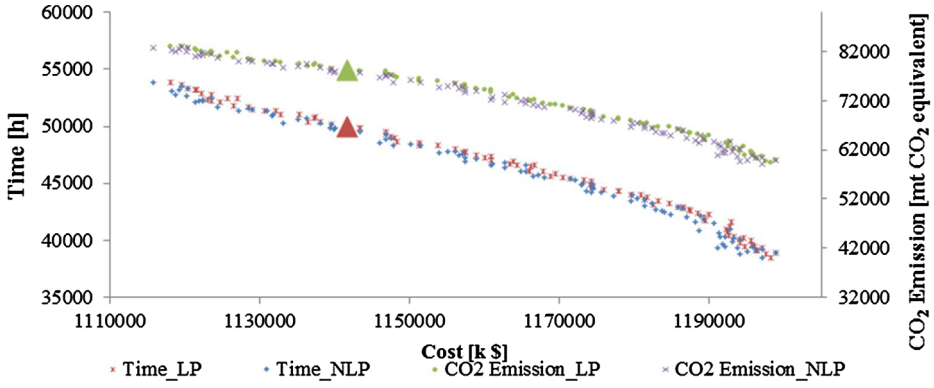


Fig. 4. Pareto solutions for the 45-node network.

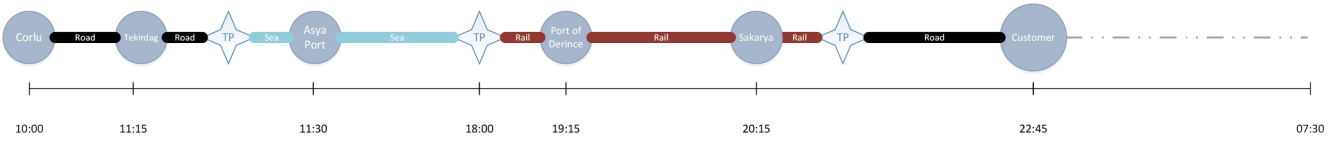


Fig. 5. Schematic representation of sample solution set.

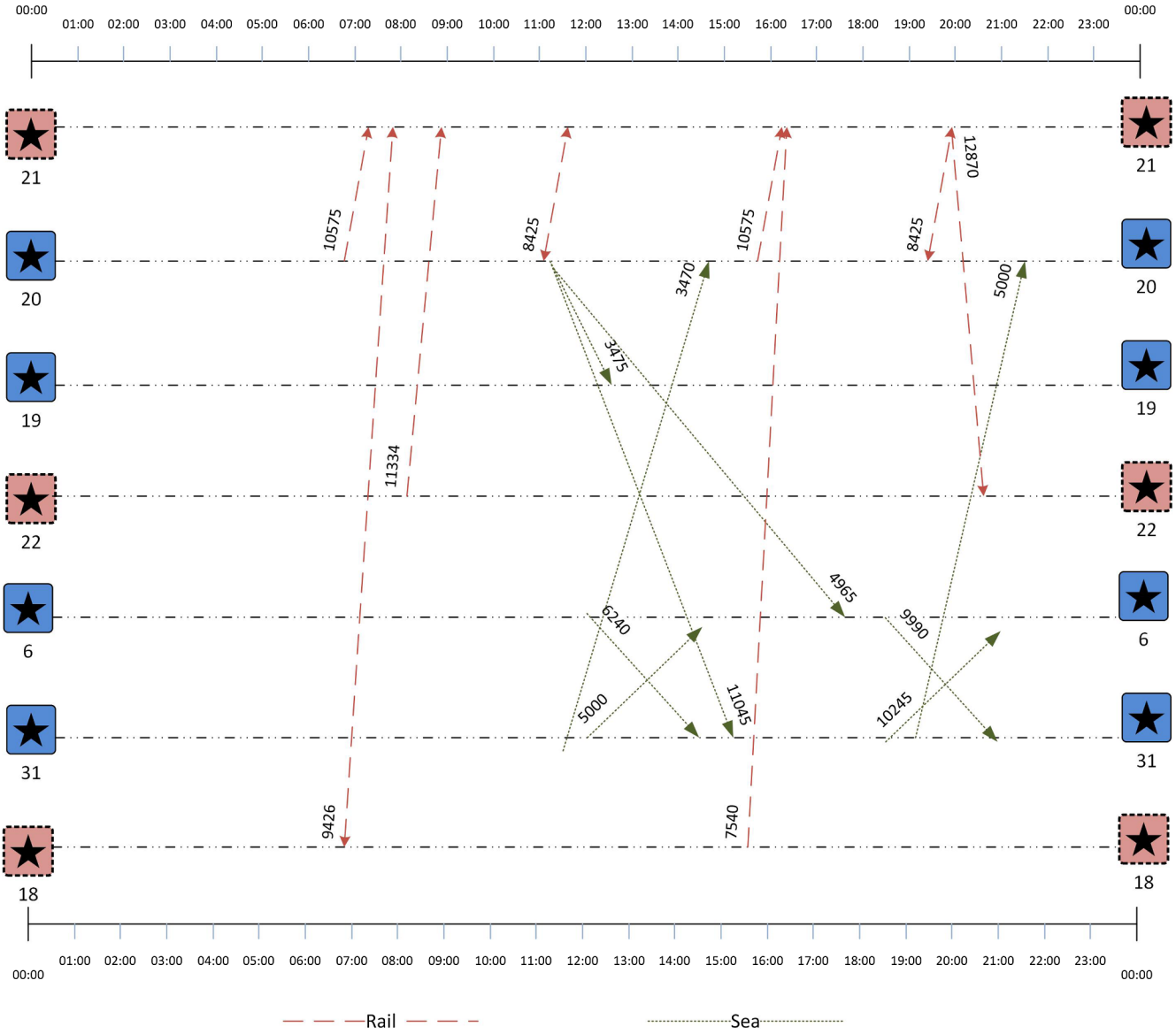


Fig. 6. Cargo movements before first disruption.

can be calculated with the BPR equation and the other modes are constant. Apart from the cost parameters, the total transportation time is another important objective of our problem. The transportation times are calculated by dividing the total distance with the average speed of the vehicles. Operational restrictions such as the opening and closing times of the ports and rail hubs and urgent delivery requests of the customers are also taken into account. For the third objective (CO_2 emissions rate), we used the same parameters given in the study by Bektaş & Laporte (2011).

The capacities of the vehicles depend on the type of the vehicles that are assumed to be either for 15 or 30-ton trucks. However, the fleet optimization model can fill up these trucks with products from different customers; this is the main difference from the intermodal transportation because only the container-unit products can be considered in intermodal transportation.

In addition, the initial demand of the customer is assumed to be known at the beginning of the day. The information with the details of sender/receiver/location can be collected, and then the model can automatically run to optimize the fleets and networks to distribute the cargo in an effective way. We consider 24 h as the planning horizon and the time interval as equal to the minimum transportation time in the entire network, which depends on the requests of the customers. The system updates its status and receives new information from the data sources at the beginning of each iteration and if necessary, the plans are revised.

4. Results

In this section, we illustrate the main concepts and analysis of the solutions on thirty-two different cases for the Marmara Region of Turkey and conduct a sensitivity analysis of the illustrative cases with additional pre-processing constraints. For each case, we analyze the effect of changing the number of the nodes present in the network, vehicle availability and upper limit for filling ratio of the vehicles. Cases include eight different network structures (between 5 and 45 nodes); two separate upper bounds on the fill ratios of the vehicles (50% and 100%) and two different ratios for the number of available vehicles (depending on the required number of vehicles for each scenario). The number of available vehicles is considered in two settings; 100% availability and 75% availability for the required capacity to fulfill the

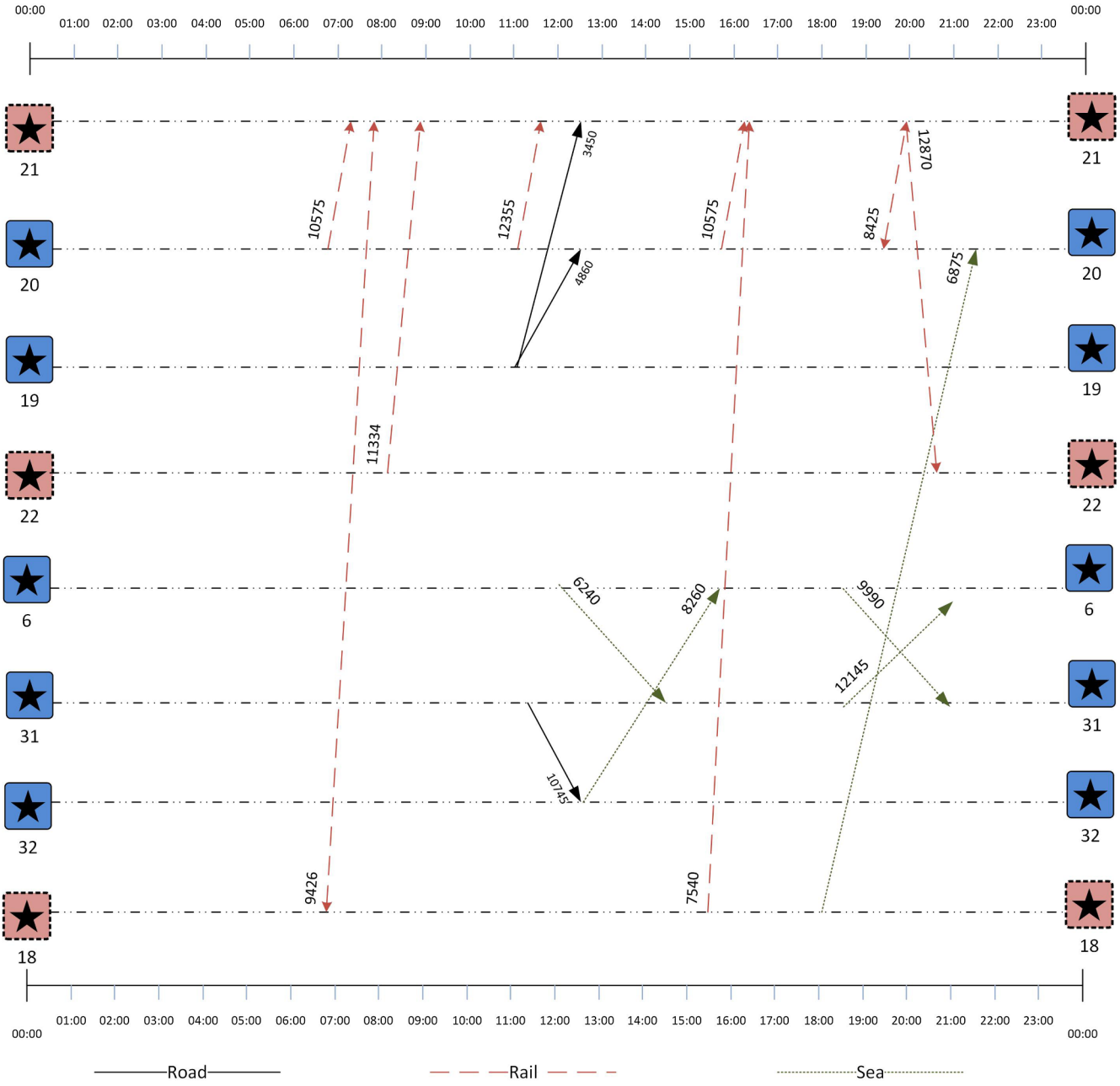


Fig. 7. Updated network after first disruption.

transport demand. This analysis shows that either the vehicles carry all of the required demand for road transportation or some of the demand is satisfied by using other types of transportation modes or increasing inventory levels in the nodes. The problem is designed and solved in two different model types, linear and non-linear; both of these models for this problem are coded in GAMS modeling environment and solved with IBM ILOG CPLEX 12.1 (CPLEX, 2009) and the non-linear one is solved by the DICOPT (Duran & Grossmann, 1986). Both problems are solved on a computer with Intel Core I5 6200 U CPU with 2.40 GHz dual core processor, and with 4.00 GB of RAM. An optimality gap of 1% is used in generating the solutions.

4.1. Illustrative case with 45-Node

We consider a number of illustrative examples consisting of different network topologies with a different number of nodes and transportation modes. The network design and transportation mode connection for the most extensive case includes 45 nodes in the network with three different transportation modes; a schematic representation of this network is given in Fig. 3. The operational information (opening, closing times, distances, etc.) is obtained from the reports of DTM (2010), KGM (2011), and TCDD (2010).

First, we solve the problem (linear and non-linear models separately) without any disruptions and 100 Pareto solutions are found for three different objective functions (cost, time, and CO_2 emissions) as given in Fig. 4.

Then, one of the Pareto solutions (marked by triangles in Fig. 4) is analyzed in detail to show the main characteristics of synchromodal transportation. This point contains solutions of three main objectives that are approximately $1.14 billion total cost, 49,962 h and equivalent to 78,248 m-tons of CO_2.

For example, the link between Corlu (Node 5) and Eskisehir (Node 26) in the highlighted solution set, the operation starts with the road transportation from Corlu (Node 5) with 15-ton truck to Asya Port,

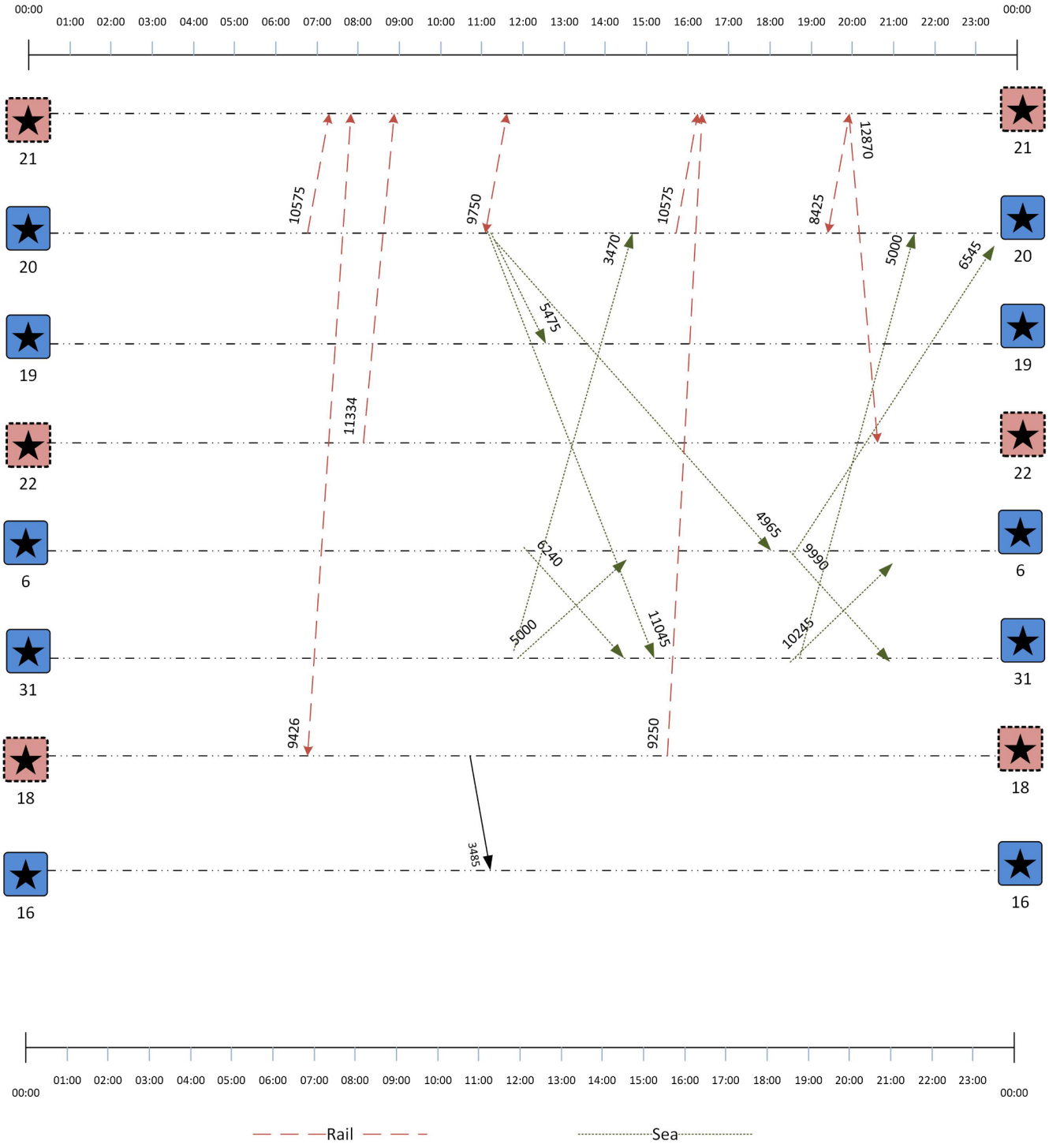


Fig. 8. Updated network design after second disruptions.

Tekirdag (Node 6). After the cargo is transferred from trucks to ferries in Asya Port (Node 6), it is carried by sea transportation and takes 6.5 h to reach Port of Derince, Izmit (Node 20). At Port of Derince (Node 20), new cargo (13 ton) is added and after consolidated into a 30-ton container and transferred to block trains, it waits approximately one hour before moving to the Sakarya (Node 22) by rail transportation. After one hour of rail transport, the cargo is taken from train to trucks at the transfer point (Sakarya (Node 22)) and it is delivered to customer by road transportation in 2,5 h. The total transportation takes approximately 12 h 45 min with a total cost of $1.200 for 26-ton cargo. However, the last-mile delivery truck should wait until the opening of the customer facility to unload the cargo and this waiting time is penalized by the system (see Fig. 5).

The main risk of synchromodal transportation is the limited schedules for train and ferry activities. The fixed schedules for these two transportation modes decrease the flexibility of synchromodal transportation. However, the model can consider this situation by sending the freight in bulk quantities (consolidation of cargo by road transportation) or keeping them in the inventory zones by considering the minimization of total cost.

An important characteristic of synchromodal transportation is its ability to react dynamically to any disruption. We consider that there

Table 2
The Average CPU Times of Linear Model with Different Combinations of Parameters [s].

Number of Nodes	5 Node				10 Node				15 Node				25 Node			
Vehicle Availability [%]	100		75		100		75		100		75		100		75	
Fill Ratios [%]	100	50	100	50	100	50	100	50	100	50	100	50	100	50	100	50
None	105	104	109	111	873	868	876	880	1316	1308	1324	1330	3452	3449	3461	3468
Eqs. (37) and (38)	102	100	104	107	858	859	862	864	1296	1295	1299	1309	3407	3398	3417	3423
Eqs. (39) and (40)	103	99	107	109	861	856	868	871	1314	1307	1315	1321	3394	3379	3402	3418
Eqs. (37)–(40)	99	97	101	107	834	824	837	841	1305	1300	1309	1313	3318	3309	3334	3352
Eqs. (37)–(43)	94	93	98	99	817	812	821	825	1286	1289	1289	1297	3254	3257	3261	3286
Number of Nodes	**30 Node**				**35 Node**				**40 Node**				**45 Node**			
Vehicle Availability [%]	100		75		100		75		100		75		100		75	
Filling Ratios [%]	100	50	100	50	100	50	100	50	100	50	100	50	100	50	100	50
None	5348	5334	5367	5413	10,084	10,079	10,112	10,201	21,287	21,194	21,342	21,675	42,087	42,009	42,223	42,351
Eqs. (37) and (38)	5314	5298	5319	5334	10,012	10,006	10,007	10,193	21,201	21,307	21,471	21,574	41,875	41,895	42,064	42,234
Eqs. (39) and (40)	5208	5205	5216	5220	9953	9967	9999	10,023	21,064	21,111	21,179	21,334	41,642	41,577	41,738	41,985
Eqs. (37)–(40)	5184	5171	5228	5231	9941	9951	9947	9974	20,945	21,034	21,196	21,234	41,547	41,487	41,584	41,821
Eqs. (37)–(43)	5161	5159	5201	5226	9912	9917	9931	9959	20,843	20,947	20,997	21,148	41,167	41,201	41,495	41,627

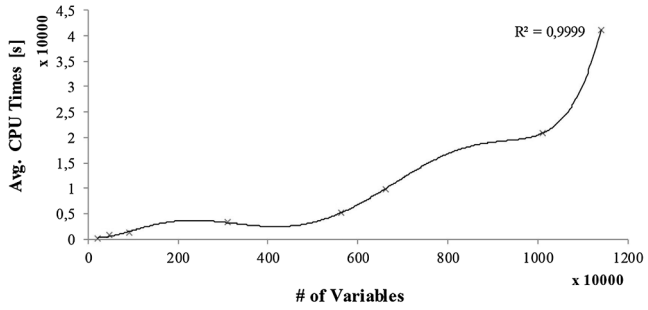


Fig. 9. Computational times vs number of variables for linear models.

are two types of disruptions during the operations;

- cancellation of the ships due to the bad weather conditions,
- cancellation of the roadways because of a traffic accident.

The information for the first disruption is registered into the system at 10:30 AM for Port of Derince (Node 20 & 21). Fig. 6 indicates the cargo movements with amount of carried cargo for rail and sea transportation during daytime around this node.

After receiving this weather forecast, the system cancels all of the sea transportation links and the expected cargo movements should be made by either road or rail transportation. In addition to sending cargo by rail or road transportation, some of the cargo waits in the inventory zones of ports to be shipped later. Therefore, the new network design after this revised plan is given and the updated transportation movements are shown in Fig. 7. After this disruption, the system gives a new solution set that includes additional road transportation from Node 19 to Node 20 and Node 21; and increased quantity of transferred cargo from Node 20 to Node 21. In addition, some of the cargo is kept in inventory and sent by sea transportation at 07:00 PM.

The second disruptive case is a traffic accident on arc (Gebze (Node 18), Port of Derince (Node 20)). This arc corresponds to one of the busiest highway segments in the Marmara Region and heavy trucks mostly use it between the industrial zones and the ports. We assume that the link is cancelled due to a traffic accident involving several cars around 10:00 AM. The cargo movements can be made by rail or sea transportation to by-pass this area. The expected time for the removal of the accident from the highway is 1.5 h and the system is revised again after receiving this information. The inventory levels are increased at the Port of Derince and Port of Haydarpasa because some trucks left their cargo in these ports to be shipped by ferries, also train capacity is full and some cargo is left in inventories in Gebze Train Station. The new planning scheme after the second disruption (in addition to the first one) is given in Fig. 8.

4.2. Sensitivity analysis

We considered thirty-two different illustrative cases, resulting in the generation of the representative set of Pareto solutions requiring a significant amount of computational effort. In order to reduce the CPU times, effectiveness of pre-processing constraints are evaluated in different combinations (see Table 2). There are three important factors that affect the effort: the number of nodes, vehicle fill ratios and availability of vehicles for road transportation. All of these factors are evaluated in various combinations. As seen in Table 2, CPU times vary dramatically with the increase in the number of nodes. For example, the 5-Node case required the least amount of computational effort with 100 s but cases with more nodes took much longer to compute, such as 42,000 s for the 45-Node case. Fig. 9 shows the relationship between the number of variables and the CPU times for cases with different number of nodes. In addition, if the system satisfies enough vehicles for the operations, the decrease in the fill ratios does not have a significant effect on the CPU times (approximately a 1% decrease). However, the 25% decrease in the vehicle availability leads to an approximately 2% increase in the CPU times. The decrease in vehicle availability and fill ratio increase the complexity and increase the CPU times of around 1–5%. The effect of this change decreases when the problem gets larger.

Also, effect of each pre-processing constraint set is the reduction of the CPU effort by approximately 1–5% for different number of nodes, however if all of the additional constraints are inserted into the models then the average CPU times decrease by around 2–10% for different cases and this reduction in computational effort increases the complexity of the problem.

As given in Fig. 9, the trend line needs power of six $O(n^6)$ with a high R^2 value. If the problem size (number of nodes, number of vehicles) gets larger, then the problem complexity increases and corresponding CPU time also increases.

5. Conclusions

In this paper, we analyze the synchromodal transportation-planning problem. In most of the developed countries, the quality of service,

reliability and efficiency of the transportation systems and supporting policy decisions are considered to be among the most important challenges. Considering the fact that transportation activities increase due to surging demand for transportation services, new ideas and concepts have emerged. Synchromodal transportation is one of these new concepts that has significant potential. The main objective of this paper is to present models and solution approaches for planning in synchromodal logistics systems and illustrate the uses of these models and solution approaches in real life cases under different scenario. The design and operation of synchromodal transportation is investigated in this paper and linear and non-linear optimization models are developed and implemented in the different cases to illustrate the characteristics of the synchromodal transportation. The performance of these models is evaluated and a through comparative analysis is conducted. Due to need of extensive computational effort and memory, we made our pilot studies in some illustrative cases. Our main objective for the future studies is to increase the complexity of the test networks and use data for a real life case in Europe. In such larger cases, we aim to develop new and more advanced exact/heuristics algorithms to generate accurate and faster solutions in our future studies.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cie.2019.03.005.

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